

• Easy • #58 • Length of Last Word

▲ **Problem Statement** • Given a string "s" containing words & spaces, return the length of the last word.

A word is a sequence of non-space characters.

▲ **Main Problem Solution** • Traverse the string "from the end"

- First, skip any trailing spaces.
- Then count characters until another space is found or the beginning of the string is ~~reached~~ reached.

This avoids creating extra strings or using extra data structures.

▲ **Brute Force Method** • Split the string into words & return the length of the last word.

Time • $O(n)$

Space • $O(n)$

▲ **Algorithm** • 1. Start from the last character

2. Skip all trailing spaces

3. Start counting characters

4. Stop when a space is encountered

5. Return the count.

▲ Important Points → Start from the end because we only need the last word.

- First skip trailing spaces.
- Then count non-space characters.
- No extra array or string is required.

▲ Complexity → Time → $O(n)$
Space → $O(1)$